

Mutual Space Generation with Redirected Walking for Asymmetric Remote Collaboration

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ABSTRACT

To enable eXtended Reality (XR) remote collaboration with co-presence, it is necessary to overcome the spatial difference between remote spaces and create a single mutual space where users can utilize their physical objects. Previous studies mainly consider semantic information and walkable area matching rate to create mutual space in a 1:1 scale, making it challenging to precisely match edge information between physical and virtual objects. In this work, we propose an edge-aware mutual space creation method by adjusting the VR client's space with Redirected Walking (RDW) technology. In our previous works, we introduced Relative Translation Gains (RTGs), a length-based RDW, and estimated the cognitive thresholds range of RTGs depending on the configuration of virtual space (size, object existence, spatial layout) to make an effect of rescaling VR clients' spaces. In addition, we introduced an RTG-based mutual space generation method that allows VR and AR users to interact with objects in physical space while they collaborate remotely. We will develop an enhanced mutual space generation method and conduct a user study, and our findings can be used to generate a mutual space for asymmetric remote collaboration.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Research on 3D avatar-based telepresence systems using Augmented Reality (AR) and Virtual Reality (VR) Head-Mounted Displays (HMDs) is being actively conducted for immersive remote collaboration. Among them, research on asymmetric collaboration situations in which remote users access the AR host's space through VR HMDs is gaining attention. This is because creating a virtual shared space based on the characteristics of the real environment and summoning remote users to the generated virtual space is relatively easy and more natural than creating an entirely new virtual environment. In order to increase co-presence between users in an AR-VR remote collaboration, we need to generate a single mutual space with a unified coordinate system from heterogeneous spaces.

Several mutual space generation methods were proposed, but most of them focus on maximizing the shared walkable area, making it challenging to utilize the physical objects in the virtual space. In this work, we propose a mutual space generation with Redirected Walking (RDW) to enable interaction with physical objects in the virtual space. We could rescale VR clients' spaces by utilizing the Relative Translation Gains (RTGs), a length-based RDW technique that can modify the VR clients' spaces without coordinate distortion. However, the RTG should be applied within a threshold range that the user does not recognize. So we investigated the effects of three components of spatial configuration (size, existence of objects, and

spatial layout) on the RTG threshold range and found the threshold range of RTGs according to each spatial configuration. Through these results, we could set a proper RTG threshold range to rescale VR clients' spaces for better spatial merging.

We proposed an RTG-based mutual space generation method based on our results that maximize the semantic match ratio between different spaces without coordinate distortion. Furthermore, based on this system, we are enhancing our algorithm using the edge-sync score, scene graph, and collaboration context. Through this, AR and VR users could interact with objects in their physical space with precisely matched edge information. Finally, we will proceed remote collaboration experiment with an object-centric XR remote collaboration scenario. Our findings will overcome spatial differences between heterogeneous remote spaces and allow immersive remote collaboration that users can utilize their physical objects in a collaborative virtual space.

2 RELATED WORKS

Redirected Walking (RDW) has been proposed to allow users in limited-sized physical spaces to explore more widely in virtual spaces accessed through VR HMD. RDW is divided into rotation gain, translation gain, and culture gain. Rotational gain and curvature gain are more effective in expanding the user's space, but there is a limitation in that the coordinate system of physical and virtual space is distorted. On the other hand, translation gain has the advantage that it could maintain the alignment between the physical and virtual space. This enables both AR and VR users could utilize physical objects such as desks and walls because the planar and edge information between physical and virtual space is matched. With these advantages, we decided to use translation gain as a space-rescaling term for the VR clients' spaces. Through this, we could secure more areas of mutual space and precisely match edge information when we generate a mutual space.

RDW utilizes the user's visual cognitive error in VR to transform the user's actual movement to the mapped movements through the HMD. Therefore, adjusting the user's movement within a threshold range is necessary to avoid increasing the VR simulator sickness score. This RDW threshold can be obtained by fitting users' responses to a psychometric function based on users' pseudo-2AFC questionnaire responses about their movements [5]. Previous research showed that RDW is influenced by various factors such as embodiment, FoV of HMDs, and the existence of distractors because the effects of RDW come from the user's visual cognition in the VR environment. Since the goal of our usage for RDW is matching between different spaces, we investigated the effect of spatial configuration on the RTG threshold.

In order to collaborate with a sense of coexistence, it is necessary to generate a mutual space by matching heterogeneous spaces. From the viewpoint of connecting spaces between spaces, there were multi-space connecting methods by connecting the most similar points between different spaces using the artificial potential fields method. It has the advantage that it could link every position of spaces, but this method often breaks the immersion when the avatar flicks or teleports to the target position. In order to ensure the free movement of users and prevent communication errors, we need to create a mutual space with the same configuration and appearance. The

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simplest way is to intersect the walkable area from each space. In addition, there is a method of matching space in the direction of matching semantic information, such as placeable and sittable areas as well as walkable areas. However, since these methods find the optimal value for maximizing the intersecting area, it is challenging to precisely match the edge information of objects or walls used for collaboration. In this work, we propose a method to precisely match the edge information of objects and walls by transforming the VR clients' space using RDW while maintaining the coordinate system.

3 RESEARCH PROGRESS

We proposed the Relative Translation Gains (RTGs), a length-based RDW technique, to provide the effect of adjusting virtual space [4]. The RTGs consist of two translation gain values to width and depth, which enables deforming the VR user's space more adaptively while maintaining the coordinate system between physical and virtual space. One of the essential characteristics of RTGs is that since it does not utilize rotational gain, the alignment of the virtual and physical space is not distorted. So after matching physical space and virtual space with RTGs, VR users can directly touch and interact with edge information from the physical desk or wall as well as walkable space. Moreover, RTGs can be utilized to match AR/VR host's space and the VR client's spaces to create a mutual space for asymmetric remote collaboration where AR and VR users connect from their own heterogeneous spaces.

Since RDW technology utilizes the user's visual cognition in VR, it is necessary to apply it within the range where users can not notice the difference. RTGs are defined as a pair of translation gains on the x-axis and y-axis, and the threshold for the RTG can be expressed as the following equation:

$$\alpha_T = \frac{g_{T,x}}{g_{T,y}} \quad (1)$$

where $g_{T,x}$ is the VR environment's x-axis translation gain and $g_{T,y}$ is the VR environment's y-axis translation gain used as the reference translation gain. As aforementioned in the Related Work section, we investigated the effects of three spatial configuration components on RTG threshold range: size, objects' existence, and spatial layout [2, 3]. We conducted two studies: Study 1 is about the size and existence of objects (3×2), and Study 2 is about the size and spatial layout (2×4). We obtained the corresponding RTG threshold range from each spatial configuration using the pseudo-Two-Alternative Forced-Choice (pseudo-2AFC) method from participants' responses [5]. Moreover, our statistical results show that all three spatial factors significantly affect the RTG threshold range. Based on our findings, we could set an appropriate RTG threshold according to the spatial configuration of the target reference space.

We proposed a simple mutual space generation method with RTGs for a table-centric asymmetric remote collaboration [1]. We first created 3D models from each space and labeled basic geometric information (floor, wall, main collaborative object, and other objects). Next, we found the optimal RTG values within the threshold range that could maximize the semantic match rate between spaces. Based on the results, we could generate a mutual space from the AR host's reference space by utilizing the spatial matched results. In this pilot system, we consider the table as the main collaborative object and define the optimal value as the RTG value that could maximize the semantic information match ratio. However, we should consider more spatial information regarding the collaboration context and define better objective functions for generating a mutual space.

4 RESEARCH PLAN

We are currently developing a spatial merging algorithm with the python optimization library, pygmo¹. From the 3D digital twin input

¹<https://esa.github.io/pygmo2/>

data of remote spaces, we extracted polygonal edge information from objects, floors, and walls in the space for the input. Then, we defined the objective function, which considered not only the semantic information match rate but also the edge-sync rate, collaboration context, and scene-graph information for computing optimal values. After setting up local space to play and visualize the mutual space, we will apply redirection gain to the VR clients in real time. This will enable VR client users could utilize their physical objects while connected to the virtual environment. For the user study, we are considering an air hockey or billiard game scenario that requires users to interact with a physical desk during collaboration. Through this, we will determine the effects of our mutual space generation method on the user's remote collaboration experience.

5 DISCUSSION

In this work, we explore how to create an RDW-based mutual space where users can directly interact with physical objects during asymmetric remote collaboration. We proposed a translation gain-based spatial deformation technique RTG to maintain the alignment between physical and virtual space and obtain an applicable RTG threshold range according to the configuration of the virtual space. In an ongoing study, we are developing a mutual space generation algorithm by matching the edge information of a real object with an object in a virtual space. In addition, we are planning to conduct a user study in our mutual space and evaluate the user's performance and experience. Finally, we want advice on the following three agendas regarding the generation of mutual spaces and the progress of user studies.

1. How can we quantify the collaboration context and scene-graph similarity between spaces and applied to mutual space creation?
2. Is the air hockey scenario where AR and VR remote users use their own desks in remote spaces an appropriate experiment scenario, and what should be evaluated by the user?
3. Does the strength of this study that users could interact with physical objects edge in a virtual space, "Edge Consistency Using Redirected Walking," have a sufficient contribution?

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